The subsystem generated by an elementary subsystem is elementary, with an explicit ordering

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Editorial note. The counted statement is the author's environment at source0.tex lines 909-910 and its proof environment opens at line 911; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

Editorial note. The source carries no label-number dump and no counter values are imposed, so the printed numbering is the wrapper's default; identity is the author label, or the pinned source position where the author gave none.

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Editorial scope. One original result of the pinned author manuscript, counted once: the generated subsystem of an elementary subsystem is elementary (author label May_26_2024_2).

Allowed standard prerequisites (editorial).

- Elementary subsystem. A is elementary in M when the two structures satisfy the same formulas with parameters in A.. It is the hypothesis and part of the conclusion.
- Generated subsystem. The subsystem generated by A is the closure of A under the partial operation.. It is the subsystem shown to be elementary.
- Ordering extending a given one. An HF-ordering of the generated subsystem can be chosen so that its closure of A is A itself.. It is the extra information the proof provides.

Definition 0.1 For $A, B \in \mathcal{C}_{\mathrm{un}}$,

$A \leq B \Leftrightarrow A \subseteq B$ and there is an HF-ordering of B where A is an initial segment.

When $A \leq B$, we say that A is **strong** in B.

Lemma 0.2 If $<$ is an HF-ordering of M, $A \subseteq M$ and $\mathrm{cl}_<(A) = A$, then $\mathrm{cl}_<(\langle A\rangle) = \langle A\rangle$.

Lemma 0.3 Let M be an unconfined STS. If $A \leq M$ (as in Definition 0.1), then $\langle A\rangle \leq M$ and there is an HF-ordering of $\langle A\rangle$ over A whose final segment $\langle A\rangle \smallsetminus A$ is well-ordered. If N is an unconfined STS and $B \leq N$ and $f : A \cong B$ is an isomorphism of partial STSs, then f can be extended to an isomorphism $f' : \langle A\rangle \cong \langle B\rangle$.

Proof. We could use Lemma 0.2 to verify the first assertion, but the following proof also gives the isomorphism. Decompose $\langle A\rangle \smallsetminus A$ into a disjoint union $\bigcup_{n<\omega} A_n$, where A_0 is the set of elements of $\langle A\rangle \smallsetminus A$ which are a product of two elements of A, and A_{n+1} is the set of elements of $\langle A\rangle \smallsetminus (A \cup A_0 \cup \ldots \cup A_n)$ which are a product of two elements of $A \cup A_0 \cup \ldots \cup A_n$. Fix an HF-ordering of M with initial segment A. There are no blocks between elements of A_0, and the only blocks of elements of A_0 with smaller elements are blocks with elements of A that justify their inclusion in A_0. Hence we can move A_0 down, just above A, and we can give A_0 a well-founded HF-ordering over A. This shows that $A \cup A_0 \leq M$. By a similar argument we can move down each A_{n+1} above $A \cup A_0 \cup \ldots \cup A_n$ and give it a well-founded HF-ordering over this set. The sequence of these well-founded orders is a well-founded HF-order of $\langle A\rangle \smallsetminus A$ which makes $\langle A\rangle$ an initial segment of the order of M. If we perform a similar decomposition $\bigcup_{n<\omega} B_n$ of $\langle B\rangle \smallsetminus B$, then we can extend f inductively to an isomorphism of partial Steiner triple systems $f_n : A \cup A_0 \cup \ldots \cup A_n \cong B \cup B_0 \cup \ldots \cup B_n$. The union of this ascending chain of isomorphisms is the required isomorphism f'. □